

# COMP3018: Set Exercises — Human-Robot Interaction (HRI)

## Cultural Differences and Probabilistic Modelling in Human-Robot Interaction

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## 1- Task (1): Cultural Differences and HRI Design

### 1.1 Cultural Differences in the Acceptance of Robots (Kaplan, 2004)

Kaplan’s 2004 identification of East-West fundamental societal divergence is rooted in the observation that “**culture affects the way technology is perceived** and, in a reciprocal manner, **technological evolution shapes culture in particular ways**” (Kaplan, 2004, p. 465); i.e., the cultural, theological, and literary narratives of each region shape the societal meta-layer (*the underlying philosophical and theological frameworks governing technological integration*) that underpins robotic acceptance.

**Western Society (The Frankenstein Syndrome).** Western culture’s creation of human-like entities has chronically been regarded suspiciously: Kaplan (2004) identifies this as the “Frankenstein Syndrome”: a culturally-filtered conviction wherein “any artificially created humanoid will necessarily turn against its creator” (p. 475). This anxiety is indeed traceable to the Judeo-Christian tradition, which posits “no place for hybrids” between divine creator and human creation (Kaplan, 2004, p. 470). The Western cultural narrative therefore frames the humanoid robot as “something else, something that should not exist” (p. 476) — a transgression against the natural order. Consequently, Western societies have historically channelled robotic development toward industrial, non-anthropomorphic applications wherein the machine remains a visible *tool* rather than a would-be social entity. Kaplan further notes the concept of “narcissistic shields” (*the psychological defence mechanisms protecting human exceptionalism*) (p. 478), whereby Westerners utilise

psychological distancing to manage the discomfort of encountering machines that blur the boundary between human and artefact.

**Eastern Society (Technology Taming and Animism).** Japanese culture, by contrast, exhibits a fundamentally different ontological stance. Kaplan (2004) traces this to the Shinto tradition of *animism*, wherein “every entity, including objects and natural elements, possesses a spirit” (pp. 467–468). This worldview dissolves the rigid Western boundary between animate and inanimate, and thus humanoid robots are not perceived as transgressions but rather as natural extensions of an already-animate material world. Moreover, Kaplan highlights the cultural mechanism of “technology taming” — a recurring historical pattern wherein initially foreign technologies (e.g., firearms, writing systems) are domesticated via integration into existing cultural frameworks such as the *kata* tradition of formalised practice (p. 470). The popular Astro Boy manga franchise (p. 466) exemplifies this domestication at the narrative level: the humanoid robot is cast not as a Frankensteinian threat but as a heroic, emotionally resonant companion. Kaplan references the Amaterasu myth (p. 469) to argue that Japanese cosmology fundamentally lacks the creator-versus-creation antagonism that underpins Western technophobia.

**Implications for HRI Design.** These divergent cultural framings have direct consequences for interaction design. Whilst Western users may prefer robots that maintain clear machine-identity markers — thereby preserving the “narcissistic shield” — Eastern users may indeed welcome anthropomorphic features that align with animistic expectations. As Lecture 4 emphasised, “every country and each culture has their own concept of proxemics” (Lecture 4), and therefore a robot’s spatial behaviour must adapt accordingly: Japanese norms demand larger personal-space buffers and non-tactile greeting protocols, whereas Mediterranean cultures tolerate closer approach distances (Lecture 4). The watchword herein is cultural relativism — no universal design policy can accommodate these fundamentally different ontological commitments.

## 1.2 African Cultural Factors Influencing HRI

Whilst Kaplan’s (2004) analysis is confined to the East-West axis, a complete account of cultural factors in HRI must address the African context, wherein distinct philosophical and socio-structural dimensions shape technology acceptance.

**Ubuntu Philosophy and Communal Identity.** The most prominent cultural factor is *Ubuntu* — the Southern African philosophical principle that “a person is a person through other persons” (Metz, 2007, p. 323). Whereas Western HRI design typically foregrounds individual user experience, Ubuntu-informed design would prioritise communal benefit and relational harmony. A robot operating within an Ubuntu-oriented society should therefore be designed to address the *group* rather than the individual, facilitating collective decision-making and shared resource access (Metz, 2007). This stands in stark contrast to the individualised personal-assistant paradigm prevalent in Western HRI.

**Power Distance and Hierarchical Norms.** Hofstede’s (2001) cultural dimensions framework identifies many African societies as exhibiting high Power Distance — a societal acceptance of hierarchical authority structures. In regard to HRI, this suggests that robots interacting with users across different social strata must modulate their behaviour accordingly: deferential language and posture when addressing elders or authority figures, and indeed a more directive interaction style when assisting in contexts where the robot is perceived as an institutional representative (Hofstede, 2001). Failing to encode these hierarchical norms risks contravening deeply held social expectations, thereby undermining trust.

**Oral Tradition and Multimodal Communication.** African cultures have historically privileged oral knowledge transmission over written documentation (Vansina, 1985). This has direct implications for interaction modality: voice-driven, narrative-based interfaces — utilising speech processing and prosodic features such as pitch and MFCCs (Lecture 3) — may achieve higher engagement than text-heavy GUI paradigms. Furthermore, as Lecture 2 established that “65% of communication is non-verbal,” gestural and paralinguistic channels become critical design considerations for African contexts wherein oral expressiveness is culturally normative.

**Infrastructure and Access Constraints.** Despite rapid technological growth, many African regions face infrastructure limitations including intermittent connectivity and limited access to high-specification

hardware (Wyche and Steinfield, 2016). HRI systems deployed in these contexts must therefore be robust to connectivity loss, operable on low-power devices, and designed for shared rather than personal ownership — aligning with the communal ethos of Ubuntu.

### 1.3 Regional Design Traits (Appearance and Behaviour)

The cultural factors identified above dictate distinct morphological and behavioural traits to maximise acceptance.

**The East (Japan).** Because Shinto animism dissolves the natural/artificial boundary (Kaplan, 2004), anthropomorphic or highly expressive aesthetic traits are welcomed. However, to align with the *kata* tradition of harmonious form, the robot’s movements must be fluid and graceful rather than purely functional. Behaviourally, the robot should adopt a “side-by-side” cooperative posture rather than an imposing face-to-face stance, reflecting Japanese non-tactile proxemic norms requiring larger personal-space buffers (Lecture 4). As Lecture 2 establishes, *haptics* (deliberate physical communication) has beneficial effects primarily within the same social group — hence physical touch is replaced by proxemic attentiveness, maintaining Hall’s personal zone (0.5–1.2m).

**The West (Europe/North America).** To avoid triggering the Frankenstein Syndrome and to respect the “narcissistic shield” (Kaplan, 2004, p. 478), Western robots should possess functional, machine-like aesthetic markers (e.g., visible joints, metallic chassis) to clearly signal their artificiality, thereby preventing descent into the uncanny valley (Mori, 1970). Behaviourally, they must exhibit extreme transparency, explicitly stating their operational reasoning to alleviate fears of autonomous transgression. The handshake serves as a *symbolic gesture* (a movement with a culturally agreed-upon meaning) combined with *haptics* (Lecture 2), albeit calibrated to the Mediterranean versus Northern European proxemic distinction (Lecture 4).

**Africa.** Informed by Ubuntu and high Power Distance (Hofstede, 2001), an African-deployed robot should possess a modest physical stature to avoid perceived challenges to human hierarchical authority. Behaviourally, it must be group-facing rather than dyadic, utilising a warm, highly expressive vocal synthesiser capable of rendering the rich prosodic variations (pitch, tone) necessary for an oral-tradition society (Lecture 3; Vansina, 1985). Furthermore, as Lecture 2 established that “65% of communication is non-verbal,” gestural and paralinguistic channels — *beat gestures* (rhythmic hand movements accentuating speech rhythm) and *iconic gestures* (movements visually representing the subject) — become critical design considerations.

### 1.4 Adapting Design Patterns for Sociality (Kahn et al., 2008)

Kahn et al. (2008) identify eight “psychological benchmarks” for social robots, noting the design space is “likely under-described” (p. 100). These patterns must be regionally adapted using the traits established in Section 1.3:

#### Pattern 1: Initial Introduction (Kahn et al., 2008, p. 100).

- *East:* Rather than tactile handshakes, the robot must initiate interaction with a calibrated bow, as “in Japan it’s very considered impolite if you break the personal distance or space and try to touch somebody” (Lecture 4). The bow angle should parametrically encode social hierarchy recognition.
- *West:* The *Initial Introduction* can incorporate the handshake as a tactile greeting, albeit with sensitivity to the Mediterranean (closer) versus Northern European (distant) proxemic distinctions (Lecture 4).
- *Africa:* Reflecting Ubuntu’s communal orientation, the introduction must address groups rather than individuals. Encoding Power Distance norms, the robot must always greet the eldest or most-senior member first (Hofstede, 2001).

#### Pattern 2: Personal Interests (Kahn et al., 2008, p. 101).

- *East:* The robot’s backstory can be elaborate, integrating seamlessly into the animistic expectation of objects possessing a “spirit” or character (Kaplan, 2004).
- *West:* Self-disclosure must be transparently mechanical, framing its “interests” around its programmed purpose to reinforce the user’s ontological comfort — supporting Theory of Mind mechanisms (Lecture 1) whereby the robot must infer the user’s mental state.

- *Africa*: This pattern should be realised via storytelling and voice-based dialogue. The robot must establish *joint attention* (the ability of multiple agents to focus on a shared reference point) via structured eye gaze to maintain narrative authority (Lecture 2; Lecture 3).

**Pattern 3: Recovering From Mistakes (Kahn et al., 2008, p. 102).**

- *East*: The robot should employ indirect acknowledgement strategies that preserve social harmony, avoiding direct self-criticism that may cause discomfort via loss of face.
- *West*: The robot should explicitly explain its error and how it will correct it, prioritising transparency to mitigate underlying technophobia.
- *Africa*: Error recovery must be strictly calibrated to social rank. An unacknowledged robot error affecting an elder constitutes a severe social violation; recovery must involve a disproportionately deferential apology to seniors, leveraging posture and movement (Lecture 2) to express contrition, thereby protecting the user’s social face within the community.

## 1.5 Summary of Cultural Design Implications

Table 1: Cultural factors and corresponding HRI design adaptations.

| Region       | Key Cultural Factor                                        | Design Adaptation                                                            |
|--------------|------------------------------------------------------------|------------------------------------------------------------------------------|
| East (Japan) | Shinto animism; non-tactile norms; <i>kata</i> tradition   | Bowing protocols; enlarged personal-space buffers; indirect error recovery   |
| West         | Frankenstein Syndrome; individualism; narcissistic shields | Machine-identity markers; handshake-ready; transparent self-disclosure       |
| Africa       | Ubuntu; high Power Distance; oral tradition                | Communal greetings; elder-first hierarchy; voice-driven narrative interfaces |

As Kahn et al. (2008) conclude, effective HRI must be “compelling as a lived experience” (p. 104) — and what constitutes a compelling experience is, as demonstrated above, inseparable from the cultural context thereof.

## 2- Task (2): POMDPs in Human-Robot Interaction

### 2.1 The Role of POMDPs in Trust, Cooperation, Coordination, and Collaboration

A Partially Observable Markov Decision Process (POMDP) extends the standard Markov Decision Process (MDP) — as studied in COMP3003 — by relaxing the assumption of full state observability. Whilst an MDP assumes the agent has direct access to the true environment state, a POMDP acknowledges that in most real-world HRI scenarios, the robot can only *infer* the true state via noisy, incomplete observations (Kaelbling, Littman and Cassandra, 1998). Formally, a POMDP is defined by the 7-tuple  $\langle S, A, T, R, \Omega, O, \gamma \rangle$ , where  $S$  is a finite set of states,  $A$  the available actions,  $T(s, a, s') = P(s' \mid s, a)$  the transition function,  $R : S \times A \rightarrow \mathbb{R}$  the reward function,  $\Omega$  a finite set of observations,  $O(s', a, o) = P(o \mid s', a)$  the observation function, and  $\gamma \in [0, 1)$  the discount factor.

To understand the POMDP’s utility, one must strictly differentiate interaction paradigms as defined in Lecture 1. **Coexistence** involves agents sharing an environment but completing different tasks, requiring only fully-observable physical states to avoid collisions. **Cooperation** involves a shared workspace and complementary tasks. However, true **collaboration** demands a shared workspace and the *exact same shared goal* (Lecture 1). In collaboration, the robot must continuously align its actions with the human’s unobservable mental states — trust, intent, cognitive load. Because the robot is fundamentally blind to these latent variables, the POMDP’s belief state  $b$  becomes the computational prerequisite for graduating from mere coexistence to true collaboration. The POMDP therefore provides the principled mathematical framework for modelling trust, coordinating joint actions under uncertainty, and enabling genuinely collaborative human-robot teams (Chen et al., 2020).

### 2.2 Uncertainty, Belief States, and Decision-Making

Because the true state is hidden, the POMDP agent maintains a **belief state**  $b$ : a probability distribution over all possible states  $S$ , where  $b(s)$  represents the agent’s subjective probability that the environment is in state  $s$ , such that  $\sum_{s \in S} b(s) = 1$ . After taking action  $a$  and receiving observation  $o$ , the belief state is updated via **Bayesian filtering**:

$$b'(s') = \eta \cdot O(s', a, o) \sum_{s \in S} T(s, a, s') \cdot b(s)$$

where  $\eta$  is a normalisation constant ensuring  $\sum_{s'} b'(s') = 1$ . This update rule captures the core epistemic challenge of HRI: the robot must continuously revise its model of the human’s internal state as new — and potentially contradictory — evidence arrives. As Lecture 3 discussed regarding affective computing, the robot utilises descriptors (e.g., pitch, MFCCs, zero-crossing rate) to extract observations from the human’s behaviour, and thus *feeds* these into the belief update process wherein each observation incrementally refines the robot’s understanding.

The belief state  $b$  therefore serves as a **sufficient statistic** for the entire interaction history — it compresses all past actions and observations into a single probability vector, enabling decision-making without storing the full trajectory (Kaelbling, Littman and Cassandra, 1998). This is indeed analogous to how a social robot must infer a human’s emotional state from noisy multimodal cues (facial expressions, vocal prosody, posture) rather than accessing the “ground truth” of their feelings directly (Lecture 1; Lecture 3).

### 2.3 Challenges of Trust Modelling and the POMDP Response

Trust is a latent psychological variable — it cannot be directly measured, only inferred from observable behavioural indicators. Three core challenges arise. First, **the measurement problem**: observations such as task compliance rate, response latency, gaze direction (Lecture 2), and verbal affirmations are noisy proxies — a user may comply with a robot’s suggestion despite low trust (e.g., due to time pressure), or indeed refuse despite high trust (e.g., due to task complexity). Second, **temporal dynamics**: trust evolves non-linearly

— it builds slowly through consistent performance but degrades rapidly after errors. The *Recovering From Mistakes* pattern (Kahn et al., 2008, p. 102) is therefore critical: a robot that acknowledges and corrects errors can arrest trust decay, whereas one that ignores failures risks appearing “aggressive” (Lecture 4). Third, **computational intractability**: solving POMDPs exactly is PSPACE-complete (Papadimitriou and Tsitsiklis, 1987), as the belief simplex is continuous even with finite  $|S|$ . Practical HRI applications therefore utilise approximate solvers — point-based methods such as PBVI (Pineau, Gordon and Thrun, 2003) or online Monte Carlo tree search such as POMCP (Silver and Veness, 2010) — to achieve tractable real-time planning.

The POMDP addresses these challenges by encoding trust as a hidden state variable, observations as probabilistic signals thereof, and actions as trust-modulating strategies (Chen et al., 2020). The belief state thereby provides the robot with a principled estimate of trust that is continuously refined, rather than a brittle threshold-based heuristic.

## 2.4 Proposed POMDP Model: Medication-Assistance Robot

To concretise this framework, I propose a POMDP model for an assistive robot supporting elderly users with medication adherence. The robot must remind the user, offer physical assistance, and calibrate its interaction based on inferred trust.

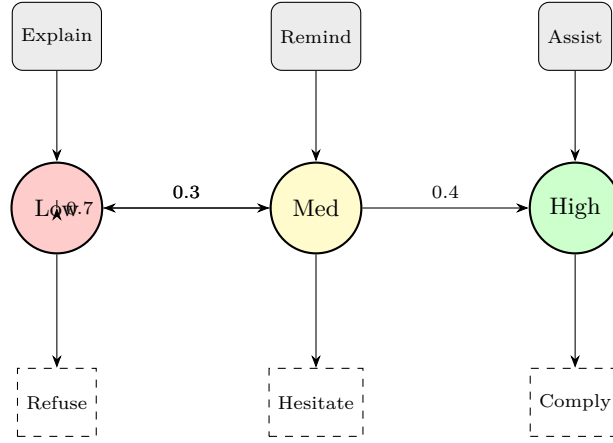


Figure 1: Simplified Trust-POMDP for an assistive medication robot. Hidden trust states (circles), noisy observations (dashed), and available actions (rounded).

### Formal Specification:

- **State Space ( $S$ ):**  $\text{Trust} \in \{\text{Low}, \text{Medium}, \text{High}\} \times \text{Task} \in \{\text{Pending}, \text{Taken}, \text{Refused}\}$ , yielding  $|S| = 9$ .
- **Action Space ( $A$ ):**  $\{\text{Verbal\_Remind}, \text{Explain\_Benefits}, \text{Offer\_Physical\_Assist}, \text{Increase\_Autonomy}, \text{Disengage}\}$ .
- **Observation Space ( $\Omega$ ):**  $\{\text{Comply}, \text{Hesitate}, \text{Verbal\_Refuse}, \text{Ignore}, \text{Gaze\_Avert}\}$ .
- **Transition Function ( $T$ ):** Models trust dynamics parametrically. If the true state is Medium Trust and the robot performs Offer\_Physical\_Assist when unneeded — thereby violating personal proxemics (Lecture 4) — trust degrades:  $P(\text{Low} \mid \text{Med}, \text{Assist}) = 0.6$ ,  $P(\text{Med} \mid \text{Med}, \text{Assist}) = 0.4$  (summing rigorously to 1.0). Conversely, a well-timed Explain\_Benefits action yields  $P(\text{High} \mid \text{Med}, \text{Explain}) = 0.5$ ,  $P(\text{Med} \mid \text{Med}, \text{Explain}) = 0.5$ .
- **Reward Function ( $R$ ):** Successful medication adherence yields  $R = +10$ ; preserving user autonomy (choosing Increase\_Autonomy when trust is High) yields a bonus  $R = +3$ ; unwanted physical assistance incurs a penalty  $R = -5$ , reflecting the social cost of proxemic violation.

**Benefits and Limitations.** The model’s strength lies in its principled handling of the trust-autonomy trade-off: the belief state  $b$  herein represents the robot’s probabilistic estimate of trust, continuously refined via

Bayesian filtering throughout the interaction wherein each observation shifts the distribution. However, the discretisation of trust into three levels introduces information loss, and calibrating the transition probabilities from real behavioural data remains a significant empirical challenge. Furthermore, the model assumes the Markov property — that trust depends only on the current state — whereas in practice, trust dynamics may depend on long interaction histories.

## 2.5 Ethical and Social Implications

The deployment of POMDPs in HRI raises profound ethical concerns. By mathematically operationalising trust, the POMDP framework transforms a once-theoretical psychological concept into a now-computable metric. However, insofar as the reward function solely prioritises task compliance, the optimal policy may learn to *exploit* the user’s trust — timing medication requests when inferred cognitive load is highest to force compliance. Therefore, the once-prohibitive computational barrier (now mitigated by approximate solvers) demands that designers explicitly encode user autonomy and informed consent into the reward structure, lest assistance erode into manipulation. Moreover, the observation function raises surveillance concerns: continuously monitoring gaze, speech prosody, and physiological signals to infer trust constitutes pervasive behavioural tracking, which must be balanced against the user’s right to privacy (Lecture 1). The ethical watchword is thus transparency — the robot must not merely *model* trust but actively *earn* it through explainable behaviour.

## References

□ COMPLETE ‘AVAILABLE AT:...’

### Task (1)

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### Task (2)

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# Appendices

## Appendix A: AI Declaration

|                       |                                                                                                                                                                                                                                                                                                                                                            |
|-----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Solo Work</b>      | <p><b>S1 - Generative AI tools have not been used for this assessment.</b></p>                                                                                                                                                                                                                                                                             |
| <b>Assisted Work</b>  | <p><b>A1 – Idea Generation and Problem Exploration</b><br/>Used to generate project ideas, explore different approaches to solving a problem, or suggest features for software or systems. Students must critically assess AI-generated suggestions and ensure their own intellectual contributions are central.</p>                                       |
|                       | <p><b>A2 - Planning &amp; Structuring Projects</b><br/>AI may help outline the structure of reports, documentation and projects. The final structure and implementation must be the student's own work.</p>                                                                                                                                                |
|                       | <p><b>A3 – Code Architecture</b><br/>AI tools maybe used to help outline code architecture (e.g. suggesting class hierarchies or module breakdowns). The final code structure must be the student's own work.</p>                                                                                                                                          |
|                       | <p><b>A4 – Research Assistance</b><br/>Used to locate and summarise relevant articles, academic papers, technical documentation, or online resources (e.g. Stack Overflow, GitHub discussions). The interpretation and integration of research into the assignment remain the student's responsibility.</p>                                                |
|                       | <p><b>A5 - Language Refinement</b><br/>Used to check grammar, refine language, improve sentence structure in documentation not code. AI should be used only to provide suggestions for improvement. Students must ensure that the documentation accurately reflects the code and is technically correct.</p>                                               |
|                       | <p><b>A6 – Code Review</b><br/>AI tools can be used to check comments within the code and to suggest improvements to code readability, structure or syntax. AI should be used only to provide suggestions for improvement. Students must ensure that the code accurately reflects their knowledge and is technically correct.</p>                          |
|                       | <p><b>A7 - Code Generation for Learning Purposes</b><br/>Used to generate example code snippets to understand syntax, explore alternative implementations, or learn new programming paradigms. Students must not submit AI-generated code as their own and must be able to explain how it works.</p>                                                       |
|                       | <p><b>A8 - Technical Guidance &amp; Debugging Support</b><br/>AI tools can be used to explain algorithms, programming concepts, or debugging strategies. Students may also help interpret error messages or suggest possible fixes. However, students must write, test, and debug their own code independently and understand all solutions submitted.</p> |
|                       | <p><b>A9 - Testing and Validation Support</b><br/>AI may assist in generating test cases, validating outputs, or suggesting edge cases for software testing. Students are responsible for designing comprehensive test plans and interpreting test results.</p>                                                                                            |
|                       | <p><b>A10 - Data Analysis and Visualization Guidance</b><br/>AI tools can help suggest ways to analyse datasets or visualize results (e.g. recommending chart types or statistical methods). Students must perform the analysis themselves and understand the implications of the results.</p>                                                             |
|                       | <p><b>A11 - Other uses not listed above</b><br/>Please specify:</p>                                                                                                                                                                                                                                                                                        |
| <b>Partnered Work</b> | <p><b>P1 - Generative AI tool usage has been used integrally for this assessment</b><br/><br/>Students can adopt approaches that are compliant with instructions in the assessment brief.<br/><br/>Please Specify:</p>                                                                                                                                     |

Figure 2: Student Declaration of AI Tool use in this Assessment Table

I declare that I've used the AI tools listed below whilst preparing this assessment. I've read and understood the University of Plymouth's policy on the use of AI tools in assessment and confirm that my use falls within the coursework's allowed categories, i.e. **A2 (Planning and Structuring Projects)** and **A4 (Research Assistance)**.

| AI Tool Used | Purpose of Use                                                                                                                       | Extent of Use                      |
|--------------|--------------------------------------------------------------------------------------------------------------------------------------|------------------------------------|
| ChatGPT      | Finding relevant pages to read in the paper ( <b>A4</b> )                                                                            | Few times if the paper is too long |
| ChatGPT      | General conversations via web-search AI about prevalent papers to read about how the topics relates to others' studies ( <b>A4</b> ) | Few times at the end               |

- ☒ I understand that the ownership and responsibility for the academic integrity of this submitted assessment falls with me, the student.
- ☒ I confirm that all details provide above are an accurate description of how AI was used for this assessment.